LIGHTYEAR

MS 94 Factory measurement and B06 engineering status

October 14 reconciliation

The retained B05 result remains 19 successes out of 20 trials, with the recorded Wilson 95 percent interval of 76.4 to 99.1 percent. B04 remains void. Subsequent B06 engineering work does not add successes to that measurement or establish a new qualified campaign.

Howard explicitly lifted the B06 merge hold on October 14 and authorized integration of the outstanding PRs followed by this documentation reconciliation. Historical references to that merge hold below and in dated source notes are superseded for publication. This instruction is not a new execution window, signed operational approval, cleanup attestation or authority to reactivate an old launcher.

Since the October 5 update below, the implementation gained resolved runtime and class provenance, observer-v2 handling, preserved failure-prefix audits, bounded observer storage, measured host/JDK alternatives and exact engineering-request preparation. The October 10 native diagnostic failed with observer Java heap exhaustion; the available trace does not prove the initiating root cause. Its failed outcome and captured evidence remain unchanged.

The October 13 R4 record describes a replacement afternoon preparation after the morning assets and authority were lost during worktree cleanup. Its published readiness observation was NOT READY because the Docker engine inventory was unavailable and free memory was below the launch minimum. The record verifies 111,311 assets at its exact execution commit. No later terminal execution or authenticated cleanup receipt is supplied by this reconciliation, so no successful October 13 native outcome is asserted. The dated window has expired and is historical.

The later engineering changes are implemented and tested within their recorded scopes. A fresh native qualification, complete measurement preflight and separate launch authority remain distinct evidence requirements. The full source inventory in the reconciliation record links the dated notes without replacing them.

Later engineering records

- Preserved native heap failure
- Bounded observer and diagnostic implementation
- Host memory measurements and superseded reproduction gate
- Retained Linux JDK alternatives
- October 13 initial engineering preparation
- Replacement R4 review
- Dated October 13 runbook

Preserved earlier milestone record

Only dash and prime character encoding has been repaired in the readable history below. The reconciliation source map retains the exact original UTF-8 text.

MS94 B05 final pass rate: 19/20 (95%); Wilson 95% interval: 76.4%-99.1%.

MS94 B05 publication and B06 preparation

October 5: B06 offline finalization and Codex account proof

This increment is delivered in two separate PRs: native integration and finalization #255 and dedicated-account Codex proof #256. Their implementation heads are `85d9dca713e22179f8b9c057 3c400c7dba87f42b` and `52a065844c690c9bcf597f7ade74565a469bfa70`, respectively. Subsequent documentation commits do not change those tested implementations. Publication and merge authorize neither native qualification nor a measurement launch.

The qualification driver integrates declared native mutations and evidence-boundary controls, preserves original captures, and finalizes each slot through archive replay and owned-resource cleanup checks. Native execution, finalization and signing share the hard deadline. Unexpected results are recorded and announced before lengthy replay; attempted, unfinalized and unstarted slots remain distinct. This is tested implementation, not a completed native qualification.

The J1 offline test connects authenticated synthetic observer streams to the B05-shaped closed `candidate-runtime-exception` payload, direct delivery into an actual signed zero-model inbox, and independent delivery replay. Support/outside origins send nothing and remain equipment-suspect. Posting attribution additionally requires the same terminal exception object and thread as the support unwind, and derives closed document labels from native document/type rows rather than a candidate trace. Wrong-document, wrong-origin, terminal-binding and cross-engine mutants fail closed. J1 business predicates are unchanged.

The dedicated-account probe **passed on October 5, 11:48:44–11:49:00 PDT**. Codex CLI 0.160.0 ran as the non-admin `lyb06builder` account. Its four child PowerShell reads bound the account SID and Codex parent PID/SID: the public positive control opened, the missing file returned `ObjectNotFound`, and both existing tools/private targets returned native `ERROR_ACCESS_DENIED` (5). The host sent only initialization and fixed `command/exec` requests, with no prompt, turn, login or model call. Codex exited 0; read-only checks confirmed the account disabled, temporary executable-specific outbound firewall rule removed, and owned processes absent. Earlier failed attempts and the canceled UAC remain preserved. No security exclusions or execution-policy changes were used.

The signed public summary has content hash `1d4f3dd36728dde0dec2915817ef816b08c605012eecc121 d814d4bb7091804c`. It binds the local signed denial file `016ac7c3097c9d0ab28e9e69 bf63a26c46e295ca3c23e02a967b2edd616130c3` and process file `3947c8fff0109bd4d3e78b90 6ec368549396ffaf6015bc74af3aa722b8152984`. The real `admit_transport` check passed for those exact files and SID. Raw observations, local paths, SID and private controls remain local. This does not admit model networking or a complete measurement transport. Re-probe the final executable tree after any bound file, runtime, account or ACL change.

Validation on the separate implementation branches: **108 offline B06 tests** for #255 and **98 offline B06 tests** for #256 passed on Windows; nine milestone documentation checks passed. The PR checks retain the exact CI results for each published head. This increment used **zero Docker commands, zero native pairs and zero model calls**. The earlier, separately approved offline compilation window is recorded in its own report; window 3's 367-second failure is now explained there without rewriting its evidence.

Core-status item	Position after this increment
1. Native adapters/references	Compiled preparation and offline integration; native qualification outstanding.
2. Posting origin and replay	Terminal binding, closed projection and delivery tested offline; synthetic fixtures confer no native credit.

Core-status item	Position after this increment
3. Qualification plans	55/41/39-slot assemblies bind all four revised source hashes; executable freeze remains blocked.
4. Builder boundary	Signed v2 Windows admission tested through Controller.launch; actual Codex process proof passed separately. Final-tree re-probe and measurement transport remain.
5. Complete preflight	Not run; requires qualified equipment and an admitted executable snapshot.

The offline audit found seven differing Maven-cache copies of `NodeTestTask` and ten of `ExecutionListenerAdapter`; the other four FRAMEWORK classes had no differing Maven-cache copies. No resolved Tycho/Surefire/OSGi classpath was saved, and Howard confirmed none is available. **No loaded JUnit copy is claimed.** That resolution and complete private per-slot input assembly still block the executable plans. A new Docker run requires explicit approval and scheduling around Maintec intake. The 135 pairs remain unrun; no slots have been replaced.

The r5 integration report lists the new assembly hashes, classpath alternatives, prospective duplicate-trace expectation and remaining blockers. The separate Codex proof PR contains the hash-only success and preserved-failure summaries. B05 evidence is frozen, B04 remains VOID, and `work/ms94`, template-r1 and J1 predicates are unchanged. **Operator review; not independent attestation.**

October 4: B06 execution admission follow-up to PR243

This increment adds the execution-admission code in five separate PRs: terminal-origin delivery #244, native control hooks #245, slot/snapshot assembly #246, signed OS-denial admission #247, and compiled lock-SQL binding #248. The implementation sequence ends at `d81a61b11a3f25752482d8a1 8e74e0bdf85ce406`; this milestone update changes documentation only.

The external JVM observer binds the terminal JUnit result to the observed throwable. A candidate-origin runtime exception can reach a signed zero-model inbox through the frozen direct diagnostic category, with offline delivery replay. J1's preflight explicitly requires that complete path. J1 integration imports its unchanged business predicates; the deliberate database failure hook targets only a declared owned container and preserves its signed intent.

Slot assembly binds private inputs, checkpoint/register and exact clock/runtime roles. New snapshot paths cannot overwrite prior attempts. The expected lock SQL must be present in the pinned class bytes, with matching class, constant-pool and posting-method hashes; a different observed statement is an evidence failure. Version-3 plans and signed authorizations bind the same journey-group window. Starting requires the full pair budget to fit; execution checks the deadline, and owned cleanup still runs after failure.

This is preparation, not completed qualification or an executable freeze. Windows security blocked the compiled AppContainer probe before its child could start. No protected read or successful OS denial was observed. Admission now requires an actual signed, hash-bound denial record instead of a boolean; the failed attempt remains preserved at content hash `9af9699d3f4d7c68c04b9618 67211e24f9fb8154a88594c414be00de29c8f438`. No security protection was disabled to get a passing result.

Validation: **85 offline B06 tests and nine milestone-documentation checks passed**. Host-only Java syntax compilation passed. There were **zero Docker commands, zero native pairs and zero model calls**. These checks do not replace target-image compilation, native execution or outside attestation.

The detailed preparation report and core status record the remaining work: actual private input/register assembly, pinned-image observer/control compilation and class catalog, remaining legacy mutation and

evidence-boundary integration, qualification finalization, and a successful OS denial probe. Core items 1–3 have additional tested code but remain unqualified; items 4–5 remain incomplete. Conversion refuses these gaps rather than issuing an executable plan with placeholders.

Assembly specification (blocked, not executable)	Pairs	Content SHA-256
J1	55	`533cc94c457fb35189e939560866d8a8a178f27bbde30409e2bf74b18f035c91`
J2	41	`bf2bf4a827e4f502cb71cccf6a436b362d3f1af385fb8e4ecc295bcaba545189`
J3	39	`407b8652079ce44bc8792f6aad4701e54a1e7813c2b02dab87f7007a9300e009`

October 17 remains the requested go/no-go, **currently no-go**. A Docker qualification run means one journey's fixed group of serial fresh pairs in an explicitly approved window. The proposed shared preparation window is October 14, 09:00–11:00 PDT: observer compile 5–15 minutes, class/Java catalog 5–15 minutes, distinct reference/control compilations 45–90 minutes. Proposed native windows are October 18–19 for J1, October 20–21 for J2, and October 22–23 for J3; estimated serial Docker time is 22.5–45 hours across the 135 pairs. The exact schedule proposal has hash `36cfc52f480c98b8b8b917397793ed3fd721031ce7d50e9084967cb0e90b1f8c`. Every window is unapproved and subject to Maintec intake availability. Publication and merge approval authorize neither Docker execution nor measurement launch.

B05 evidence remains frozen, B04 remains VOID, and `work/ms94`, template-r1 and J1 predicate files are unchanged. Earlier failures remain preserved. Operator review, not independent attestation.

October 4: B06 PR241 review fixes and qualification drafts

The review report records implementation commit `b858e0d05a28cac41ab7b1a759fbdc827b27e888`. Missing or malformed candidate traces and candidate timeouts now have typed candidate results. Business rejection records bind the entry, clock and capture evidence and distinguish completed gate stages from stages never reached. Clock samples and runtime container identities are checked against the plan.

The external posting collector and independent cause replay have synthetic Oracle/PostgreSQL controls for prior-post, prior-lock, support-origin, outside-origin, wrong-document and genuine equipment faults. Synthetic fixtures are not native evidence and do not qualify diagnostic forwarding.

Validation: **68 offline B06 tests passed**, plus host-only Java syntax compilation. This increment made **zero model calls, zero Docker runs and zero native pairs**. The validation manifest has content hash `59cba00d08683012c0c452bb15d0091e10e4a22125179b67db33e892fb055e9b`.

Review draft	Fresh pairs	Plan content SHA-256
J1 requalification	55	`2bffc06197d159cc74f772c4e1fa6cf3c4c86d0d3fe304cab2122a957e9266da`
J2 procure-to-pay	41	`0e49063add148f0c15b7346235acc54732f988e155f991e81140c012f321923e`

Review draft	Fresh pairs	Plan content SHA-256
J3 materials	39	`9456b8448cfd08caba106efa5d925c63275c35ce91b89e73f8323768f4dd1e1d`

The 135-pair planning estimate is **22.5–45 hours of serial Docker work**, excluding compilation and offline replay; observer overhead remains unmeasured. These are non-executable review drafts. Native hooks, target-image compilation, per-slot input/catalog bindings and production replay/delivery integration remain pending. The public-tool boundary rejects filesystem reads, but actual OS-level denial of builder access to `tools/` is **not yet confirmed** and remains an admission gate.

Against core-status items 1–5: items 1–2 have tested preparation, item 3 has review drafts rather than an executable freeze, and items 4–5 remain incomplete. Publication grants no qualification or measurement launch authority. Every Docker run still needs explicit approval and must not overlap the October 5 CardDemo intake. B05, work/ms94, template-r1 and J1 implementation files are unchanged; B04 remains VOID. Operator review, not independent attestation.

October 4: B06 native admission preparation

The native adapter increment was merged through PR #241 at `538c0829766ca3d0d0611a46f7a06bbbaba2eba5`, with all 25 CI checks passing. Its exact published head was `fb2341d20f85369c437e57c52b2e13852102bc9b`. The adapter report binds full-entry and all-table reconciliation code, private J2/J3 reference hashes, typed failures, real-clock evidence and owned cleanup. J1 predicates remain unchanged. The private reference sources and expectations were not uploaded.

The separate observer preparation report documents the external JDI collector, suspended native SQL readbacks, host-written event chain and offline integrity replay. It also records the remaining causal attribution work: **collection is not qualification and does not authorize diagnostic forwarding**. An early error cannot claim completed entry replay.

Validation for the combined preparation: 50 B06 Python tests passed on Windows. The J2/J3 reference compilation checks took 111.969 seconds; three preserved observer compilation checks took 9.625 seconds. Each used the pinned image with no network or database execution, and each owned compiler container was verified absent. The current observer source hash is `b55882fdd8c9b8d71d1bf927a9940982a4508dac41249069a805dfe036e88632`; its latest compilation record is `d4f2443391d156e2dd5010569600231ab996e77f529add0a6a5f214047e770bf`. Overall preparation wall time was not measured. These costs are not measurement or native qualification costs. Zero native pairs, JVM attachments and model calls were performed by these increments.

Against core-status items 1-5: adapter/reference code is present but native acceptance and remaining scope/register admission are pending; observer collection/replay is present but causal proof and native controls are pending; exact qualification plans are not frozen; the full measurement executable and supervised preflight remain incomplete. The next plan commit requires Howard's approval before native qualification. Docker work must not overlap the October 5 CardDemo intake and requires explicit permission.

B05 evidence, work/ms94 and template-r1 were not changed. B04 remains VOID. This milestone records operator review, not independent attestation, and grants no B06 launch authority. Earlier milestone results and failures below are preserved.

Earlier publication and preparation record

Updated October 3, 2026. B05 is closed and frozen. B04 remains VOID. B06 preparation is implemented and tested in the scope below; native qualification and the full zero-model preflight have not passed. No B06 model calls are authorized.

B05 publication completed

The terminal audit and results are on main through PR #234. The preregistration and executable/preflight evidence are now also on main: PR #236 merged the plan branch at `18a748d81b0c11953cffc91142d77743b0b15593`, and PR #237 merged the r2 branch at `2defbb7074d453b1dd3a8c5e544cae2fa08297e5`. Both original branch heads were verified as ancestors of public main. The r2 executable evidence and controller files retain their original bytes. PR #238 updated the results publication status and added the archive proposal at `51bf5f83c77d5a187deda9d6f6963dab18ac299f`. All three PRs passed their checks. A read-only check still verifies the B05 frozen snapshot manifest.

B05 first-try results remain 12/20 (60%; Wilson 95% 38.7%-78.1%); final results remain 19/20 (95%; Wilson 95% 76.4%-99.1%). Three pilots are excluded. The cohort-18 support-origin halt remains a failed verdict with an operator Category C classification. No result is rewritten by this publication or the B06 work. B03 remains separate at 12/20 first try; B02 remains 2/10 (Wilson 5.7%-51.0%), with all eleven failures and five Category B public-contract caveats preserved. B01 invalid-provenance costs remain separate. There is no pooling.

Archive upload approved and clearance pending

The release plan and inventory cover 31 original signed archives totaling 3,529,643,030 bytes (3.53 GB). The proposed location is a GitHub release with one ZIP per attempt, plus the signed receipts, manifest, trusted public key and offline replay instructions. Howard approved uploading after the proposal and findings were reported. This is operator review, not independent attestation or a signed Tower decision.

No archives have been uploaded. A read-only scan found populated password fields in native captures; their public-demo provenance is not established. Disclosure review, independent scanning and a public-only clean-room replay remain pending. The upload approval is recorded, but it does not itself clear those findings. Original signed archives must not be silently redacted or re-signed.

B06 preparation delivered

The preparation status records the implemented scope and remaining admission work. The generic builder template and closed diagnostic policy were signed and frozen before the materials work order was written. J3 remains hash-only publicly; its work order and expected values are private. A preserved pre-execution draft was revised after read-only seed inspection, without candidate output or changes to the generic template.

Preparation now includes the deterministic three-journey schedule, per-journey Wilson analysis, real-clock period guard, controller core, J2 invoice/date checks, J3 quantity/cost/valuation/accounting predicates, and a fail-closed cohort-18 attribution predicate. Real Tower integration tests cover signed launch, repeated cause, equipment-suspect pause, continue, wrong-pause refusal and void without rewriting verdicts. Native posting provenance must still be implemented and qualified; a signed synthetic fixture is not native evidence.

There are six excluded pilots and 72 planned cohort slots, 24 per journey. Every journey must reach at least 21/24 final passes for the cross-journey headline; J3 always counts. Limits are 390 calls, 234 compilations and 96 hours including pauses; each trial has five calls, three compilations and 7,190 seconds including finalization. These are preparation constraints, not launch authority.

Validation: 43 Windows preparation/Tower tests and nine milestone-documentation checks passed. The real concurrent Windows status writer/reader is included. This increment used zero model calls, Java compilations or native qualification pairs. The full executable has not been frozen or preflighted for measurement.

Next gates are native J2/J3 references and adapters, the trusted posting observer, fresh qualification and terminal audits for all three journeys, complete worker and replay integration, and the supervised three-journey zero-model preflight. Preregistration and a launch runbook follow those gates. No measurement can start without Howard's later commit-specific approval and verified Tower authorization.

Preserved historical milestone through October 1

The record below is retained verbatim as historical context. Its statements that B04 had not started describe the earlier preparation state, superseded by the current VOID status above. Earlier failures, costs and caveats are unchanged.

95% Wilson interval: 38.7%–78.1%; Stage B 03: 12/20 first-try passes (60%).

MS94 Qualify the equipment, then measure the factory

Status: Stage A accepted and published; Stage B 03 complete and independently replayed; operator review accepted; equipment-06 A1 passed and independently replayed. A2 stopped on an unmet control expectation. Howard authorized A2 revision 1, subsequent A3, and gated B04; A2 revision 1 passed all 18 controls and independently replayed; The corrected A3 checkpoint preparation passed and independently replayed; paired A3 qualification stopped at its first archive on an invalid entry-admission signature; independent replay reproduced the failure. A3 revision 1 corrected signing but stopped on the perturbed retained reference with an accounting-period cache mismatch; independent replay reproduced both results. The explicit order-accounting-date sources passed all 28 fresh outcomes and independent replay; fresh paired A3 passed all fourteen intended outcomes and independent replay; B04 implementation is prepared as one new 23-slot snapshot for prospective publication; no B04 model call has occurred. Stage B 01/02 outcomes remain separate. Date: 01 October 2026 UTC (paired A3 passed; B04 prospective freeze; B03 unchanged).

On September 30 Howard accepted the four next steps: revise the failed control, run a fresh qualification, perform native A3 value-invariance, then launch B04 only after all prerequisites pass. A2 revision 1 uses a missing rollback stimulus plus omitted wait, with 18 fresh pairs and no model calls. It preserves the failed wait-only control and all earlier costs. It does not claim that omission of a wait alone was qualified. The judge, observer, diagnostic exporter and public support remain unchanged. The original A3 stopped on its first retained-reference archive; subsequent campaigns and the newly passed dated A3 are recorded separately below. B04 has not started.

MS94 first checks whether the judge and execution environment can reliably accept correct journeys and reject declared faults. Only then can a later campaign measure generated code. These human-reviewed references are test equipment; none of their passes count as autonomous factory successes.

Stage B 04 preparation

Howard signed acceptance of all four B03 review decisions. The new equipment-06 A′1 plan completed all 30 fresh, zero-model judge recheck pairs with the expected outcomes. All 30 signed native archives independently replayed and actual owned Docker cleanup was verified. The retained reference passed 10/10, giving a one-sided 95% upper false-rejection bound of 25.9% (26% rounded). Revised A′2 diagnostic qualification passed; fresh dated-source A′3 private-value invariance passed; B04 still requires its own prospective public freeze. No B04 model execution has started.

A2 revision 1 passed all 18 fresh controls. All 18 archives independently replayed with the unchanged frozen gate and diagnostic verifier; all 162 owned Docker resources and six offline compilation containers were verified absent. Campaign elapsed time was 6,566.890 seconds, with zero model calls. Independent terminal verification took 411.485 seconds; offline preparation took 263.578 seconds and six compilations. The stronger missing-stimulus-and-wait control passed 3/3; this does not qualify the original wait-only fault. All costs and outcomes are separate from the preserved original A2 failure. The subsequent dated-source A3 result is reported below; B04 remains unstarted.

A′2 stopped on control 13: omitting the rollback observation wait produced a native `passed` verdict and no diagnostic instead of the declared contract violation. Both observers nevertheless recorded rollback and lock-wait witnesses. The first 12 controls met their expectations; five slots remain unstarted. A2 qualification failed and the follow-up was paused at that time. Howard subsequently authorized the fresh revision; the later A3 result is reported below and B04 remains unstarted. All 13 signed archives independently replayed with the unchanged gate and diagnostic verifier, and all 117 owned Docker resources were verified absent. Campaign elapsed time was 4,829.547 seconds, with zero model calls. Preparation and replay costs are reported separately in the linked result. No frozen result was amended or reinterpreted.

A3 checkpoint preparation stopped before checkpoint admission: the new native catalog lineage fields did not match the unchanged schema verifier's required MS84 checkpoint binding. Both native transform audits and 60 constraint probes completed, but this does not qualify A3. Independent evidence replay reproduced the same exception, and actual Docker inventory confirmed all five owned resources absent. Preparation took 215.641 seconds; verification took 8.859 seconds. There were zero candidate executions, Java compilations or model calls. The failure is preserved, the follow-up is paused, and B04 remains blocked. Any correction requires a fresh prospective preparation rather than changing this frozen run.

Howard authorized correcting the checkpoint binding and running A3. The fresh corrected preparation admitted a genuinely perturbed private checkpoint in 269.906 seconds. Independent replay took 27.734 seconds and verified all five owned resources absent. Zero candidate executions, Java compilations or model calls were used. The earlier preparation failure remains unchanged; its pause was superseded by Howard's explicit resumption.

The A3 native qualification plan now declares 14 fresh pairs: the retained reference and all six revised faults, each against the original and derived checkpoint. It requires exact diagnostic byte equality, full native entry admission, independent archive replay and actual cleanup, with zero model calls, an eight-hour cap and no replacements or restarts. A3 stopped on its first archive after the admitted-checkpoint retained reference passed natively. Its entry record was sealed before signing, causing the envelope verifier to reject the signature. Unchanged frozen archive replay independently reproduced that failure; this is an equipment evidence defect. Thirteen slots remain unstarted, and no perturbed qualification run or diagnostic pair comparison completed. Campaign elapsed time was 467.547 seconds with zero model calls. Independent terminal verification took 29.109 seconds and confirmed all nine owned Docker resources absent. All frozen sources and evidence remain unchanged. This failure paused the follow-up at that time. The subsequently authorized fresh campaigns are reported below. The checkpoint preparation, A1, revised A2 and earlier failures retain their separate results and costs.

Howard approved correcting the signing boundary and freezing a fresh campaign. A3 revision 1 signs the unsealed entry body, verifies the saved envelope before execution, and repeats all fourteen original slots under a new public freeze. The 24 focused checks passed; they are preparation evidence only. All 911 inherited files, including the failed original implementation and unchanged archive verifier, remain byte-identical. The earlier failure remains final; its pause is superseded only for this newly authorized revision. A3 revision 1 stopped after two slots: the admitted retained reference passed, but the perturbed retained reference failed the unchanged accounting-period cache predicate. Twelve fault slots remain unstarted. Both archives independently replayed full entry, complete gates and diagnostics; all eighteen owned resources were verified absent. Signing now verifies correctly. The retained pair's empty diagnostics matched, but this does not qualify the failed business reference or

any unrun fault pair. Campaign time was 1,053.453 seconds, independent terminal verification 119.719 seconds, with zero model calls. Read-only diagnosis found an October cache against September postings after crossing the UTC month boundary; a wall-clock dependency is a hypothesis to investigate, not an established perturbation effect. The signed business-failure result remains unchanged. That failure paused the follow-up; Howard subsequently authorized the new dated-source qualification below. B04 remains blocked.

Read-only follow-up established the clock-dependent path in the pinned native source: the reference omits order `DateAcct`, `MOrder` defaults it from the real JVM clock, and completion validates that date through `MPeriod`, updating the current-period cache. Native captures show September 30 order accounting dates in the admitted run and October 1 in the perturbed run, while all matching postings remain September 26. The correction now explicitly dates the order in fresh reference/control copies and requires new qualification, declared below. The signed clock diagnosis is read-only, with zero new native executions, compilations or models. It does not reinterpret the failed result or establish a perturbation-only causal effect.

Howard authorized continuing with the explicit order-accounting-date correction. The new dated-source qualification adds one accounting-date assignment to new copies of the reference and all six fault controls. Ten reference runs and eighteen fault runs are prospectively declared on the admitted checkpoint. All seven sources compiled offline and 33 focused tests passed. Native qualification then passed 28/28 intended outcomes: ten retained-reference passes, six contract violations and twelve execution failures, including three empty-feedback support throws marked equipment-suspect. All 28 archives independently replayed full entry, complete gates and diagnostics; all 252 campaign resources plus seven preparation containers were verified absent. Campaign elapsed time was 11,688.531 seconds, separate from 313.313 seconds and seven preparation compilations and 1,403.922 seconds independent replay. Model calls were zero. The new dated-reference 10/10 one-sided 95% upper false-rejection bound is 25.9%, reported separately from historical A1. Qualification is bounded to actual October 1 UTC execution dates, with real-time application clock and period-window policies unchanged. All 918 inherited snapshot files and earlier outcomes remain unchanged. After fully verified success, the fresh paired A3 below was performed. This source correction changes no builder prompt, public contract, support, judge, diagnostic exporter or clock.

The fresh paired A3 with dated sources is prospectively frozen for fourteen fresh Oracle/PostgreSQL pairs: each of the seven qualified dated sources runs admitted then perturbed. The unchanged native runner requires full entry admission before every candidate, genuine SQL restoration proofs, complete retained-reference business passes and equal actual canonical diagnostic bytes, suspicion flags and dispositions across all seven pairs. The eight-hour, zero-model campaign permits no restarts or replacements. All 940 inherited files remain unchanged. The new campaign passed all 14 intended outcomes: both complete retained-reference passes, four contract violations and eight execution failures. All seven actual diagnostic byte, suspicion and disposition comparisons matched. All fourteen archives independently replayed full entry, complete gates and diagnostics, including original/derived multisets and real per-cell restoration proofs for every perturbed slot. All 126 owned resources plus seven earlier compilation containers were verified absent before and after replay, and all 946 frozen files remained unchanged. Campaign elapsed was 5,998.484 seconds; independent replay took 594.407 seconds. No new compilations or model calls were needed. Qualification is bounded to actual October 1 UTC execution dates and the declared limited perturbation. The original WAIT-ONLY scope remains unqualified. B04 may proceed to implementation and its own exact prospective public freeze, with no model generation before that freeze. Earlier failures and costs stay separate.

Stage B 03 terminal result

Post-run classification is complete after Howard authorized analysis to resume: all eight cohort failures and pilot-01 are Category A nonempty-accounting assertions on reversal allocations. Pilot-03 is Category C, a support-wrapper throw whose underlying evidence points to candidate inventory-posting sequence misuse; its measurement-validity adjudication was accepted by Howard. All thirteen passes

are assessed as accepted with notes about fixture-dependent selection. The completed operator review packet was accepted in all four decisions with Howard Weale's typed signature. This is operator review, not independent review; the B04 classification prerequisite is satisfied. The 12/20 headline and every signed result remain unchanged; no new model call or native execution was made.

All 20 fresh cohort trials and all three excluded pilots finished. Twelve cohort candidates passed on their first native attempt; eight ended in execution failure with no supported repair diagnostic. There were no repaired passes, invalids, unstarted slots, judge errors or insufficient-evidence outcomes. The signed report is nonvoid. The three qualification controls are not included in this rate.

Native class / disposition	Cohort	Pilots (excluded)
Passed	12	1
Business failure	0	0
Execution failure	8	2
Judge error	0	0
Insufficient evidence	0	0
First-try / repaired passes	12 / 0	1 / 0
Halted with no supported repair diagnostic	8	2
Invalid / unstarted	0 / 0	0 / 0

Consumption	Cohort	Pilots	Total
Builder / analyst calls	20 / 0	3 / 0	23 / 0
Compilations / native pairs	22 / 20	3 / 3	25 / 23
Input tokens (includes cached input)	6,883,993	1,167,069	8,051,062
Cached input tokens	6,014,976	1,031,040	7,046,016
Output tokens	251,914	39,915	291,829
Agent seconds	8,829.203	1,379.172	10,208.375
Native seconds	7,005.546	1,016.781	8,022.327

Campaign elapsed time was 20,098.890 seconds (5h 34m 58.890s), including publication overhead. Independent replay then took 933.828 seconds, with zero additional model calls or native executions. The unchanged frozen `tools.ms94_publication_b_v5` reproduced all 23 complete gates and trial provenance. All signed cleanup records passed; actual Docker inventory confirmed all 207 campaign-owned resources absent (161 containers, 23 networks, 23 volumes). The controller exited after writing the signed terminal report; no campaign error or stopping record appeared. The separate process/compilation cleanup record also confirms no remaining offline compilation containers.

The frozen decision is **review the intermediate result before any further generation**. No purchasing run, ablation, restart or replacement slot was launched. The single intervention was the approved public order-derived invoice-type retention rule and matching acceptance predicate; no smoke tool, diagnostic widening or support change.

This is one baseline operations journey, not a claim about unfamiliar journeys or production qualification. Mechanical replay does not substitute for human review of the new candidates. The prior 2/10 Stage B 02 result has a different public contract and five contract-gap rejections, so these samples are not pooled and no controlled causal improvement is claimed. Its 95% Wilson interval remains 5.7% –51.0%, with all eleven original failures preserved. Stage B 01 remains invalid-provenance and void, with its one call and 459.047 seconds kept separate.

Requested model: `gpt-6-astra`, high reasoning, through the pinned signed-in CLI; provider-resolved model and per-call billed dollars are not exposed. Generation is nonzero as reported above. No GCP or organization API was used; local compute and electricity were not metered. Full native archives remain local. Only signed result/call/native/publication metadata, cleanup records and the readable summary are published here; no captures, candidate source or secrets are included.

Signed report content SHA-256: `ba30e83bb2f29a5bc4e31927a245601e170848f3d533c8e5 7cf23eb803b49ec1`. Independent terminal verification content SHA-256: `00f846bad1407fc970ee057e abee5633a870a4d923f6c754bd64c8455f5f0780`.

Stage B 03 signed report, usage summary and independent verification.

Stage B 02 result, as recorded

Stage B 02: 2/10 first-try passes (20%); 95% Wilson interval 5.7%–51.0%. Passes and six execution failures human-accepted; five comparison rejections retain a public-contract caveat.

Ten fresh operations cohort trials produced two first-try passes, eight failures and no repairs. Three preceding pilots all failed and are excluded from the rate. The 95% Wilson interval spans 45.3 percentage points; this small, single-journey sample does not establish broad autonomous development performance.

Native result class / disposition	Cohort	Pilots (excluded)
Passed	2	0
Business failure	4	1
Execution failure	4	2
Judge error	0	0
Insufficient evidence	0	0
First-try / repaired passes	2 / 0	0 / 0
Halted with no supported repair diagnostic	8	3
Invalid / unstarted	0 / 0	0 / 0

Cohort-05 and cohort-06 are the two passes. All eleven failures exported empty closed diagnostics, so no analyst repair was attempted. The frozen report is non-void and retains exactly these outcomes. Mechanical replay is distinct from human judgment of whether the acceptance contract was appropriate.

Consumption	Cohort	Pilots	Total
Builder / analyst calls	10 / 0	3 / 0	13 / 0
Compilations / native pairs	10 / 10	3 / 3	13 / 13

Consumption	Cohort	Pilots	Total
Input tokens (includes cached input)	3,416,894	1,101,948	4,518,842
Cached input tokens	3,011,072	975,872	3,986,944
Output tokens	123,653	39,420	163,073
Agent seconds	4,392.392	1,383.015	5,775.407
Native seconds	3,784.517	1,090.109	4,874.626

Campaign elapsed time was 11,738.985 seconds (3h 15m 39s), including publication overhead. Requested model: `gpt-6-astra`, high reasoning, through the pinned signed-in CLI. No GCP resources or organization API were used. Generation was nonzero, as above. Per-call billed dollars and provider-internal retries are not exposed; local compute and electricity are not metered.

The unchanged `tools.ms94_publication_b_v4` replayed all 13 complete archived gates offline in another 497.699 seconds, with zero new model calls/native executions. The initial sandbox replay could not create its temporary directories; its errors are preserved separately and were not candidate or judge outcomes. The subsequent permitted replay verified all thirteen. All signed cleanup records verified; the terminal Docker inventory found none of the campaign's 91 containers, 13 networks or 13 volumes, and no remaining compilation containers. The controller and launcher exited normally. Monitoring is stopped.

Signed report content SHA-256: `9657ade08a907ace39b382907c82edca6152f631f16fbf7b5c5c6910a2f94673`. The report remains under the frozen root at `work/ms94/stage-b-02/report.json`. The development replay receipt is `work/ms94/stage-b-02-terminal-replay-02.json`. This publication contains the result and a value-redacted classification index; it does not claim that the full Stage B native archives are publicly hosted.

Classification determines the next change

All eleven failure classifications record stage, exception/field, cause, judge assessment and diagnostic usefulness. Six candidates crash on their own incorrect requirement that a posted credit/reversal allocation have accounting rows. The pinned application deliberately permits an empty facts list for that reversal allocation. These are candidate logic errors.

Five candidates finish both lanes but replace the order-derived invoice type with an ARI API selection whose tied ordering differs between engines. The public shapes pass and monetary fields are absent from the unresolved difference set. The comparator correctly detects the different type and propagated numbering, but the public contract does not explicitly settle allowable invoice-type choices. These five are flagged for possible contract/equipment defect, not asserted to be five proven business-logic errors. No failed slot is relabelled passed.

Howard Weale accepted both passes and confirmed all six Category A execution failures on 29 September 2026. He approved adding deterministic invoice-type retention **for the next run**. That approval does not establish that it was already in the frozen builder-visible contract or retroactively justify the five rejections. The exact review statements and candidate hashes are preserved separately. Any confirmed wrong judge requires a separate measurement-validity adjudication and equipment requalification before another measurement. The original report and costs remain preserved even if its interpretation is later invalidated.

The approved prospective contract rule is the single next intervention: retain the original invoice type derived from the order, with a matching structural acceptance rule. Integrate and requalify that boundary before measuring. The provisional smoke-run tool and diagnostic widening are deferred; neither is combined with this contract change. The next measurement is **20 fresh cohort trials**, with

separately declared pilots, one fixed intervention and a publicly timestamped plan before generation. Existing trials were not reused or pooled. The subsequently authorized Stage B 03 is recorded separately above.

Accepted Stage A result

Equipment-04 ran 62 fresh Oracle/PostgreSQL database pairs with a frozen judge, observer, references, support code, schedule and 15-hour limit. It completed without an unexpected failure or stopping-rule trigger.

Check	Verified result
Operations reference	10/10 passes
Procure-to-pay reference	10/10 passes
Deliberate business mutations	39/39 rejected for the intended reason
Duplicate-trace support controls	3/3 expected execution-failure rejections
Unexpected failures / unstarted slots	0 / 0
Full evidence publications	62/62 independently replayed offline
Native execution and in-run publication	27,842.281 seconds (7h 44m)
Additional terminal publication replay	2,936.994 seconds (49m)
Builder / analyst calls	0 / 0
Cloud resources	None; local Docker only

The mutation score is 100% for the 39 declared scored cases. Duplicate-trace controls are reported separately and do not inflate that denominator. Both positive references had zero measured false rejections in their ten repetitions. This is a bounded observation, not proof of a zero population failure rate.

All 62 signed cleanup records verified. A final Docker inventory found no campaign containers, networks or volumes. Howard Weale's signed support review, independent purchasing review (non-author), and owned comparison-scope approval were validated against the exact approved source hashes. The unchanged acceptance validator recorded Stage A acceptance. Earlier failed declarations remain intact.

What changed in the equipment

The public contracts state document shapes, required links, trace fields and transaction boundaries. Operations uses one aggregate order-linked invoice line; two shipment-linked invoice lines are not silently accepted. Monetary answers remain private. Purchasing side effects are bounded by source-derived ownership, organization, accounting history and posting rules, rather than a table whitelist.

The versioned SDK supplies typed accessors, strict trace serialization, cleanup and explicit posting guards. Its guard does not prevent application-internal costing or reposting; that is an explicit reviewed limit. Unknown judge exceptions remain inspectable `judge-error` results. Business failures, execution failures, insufficient evidence and passes remain separate classes.

Equipment-04 uses `native-transaction-observer-v4` and `idempiere-qualified-judge-v5-observer-ms94`. Oracle occasionally supplied an incomplete counter tuple before supplying a complete tuple for the same session and transaction while the target write lock was still held. Observer v4 can replace that incomplete baseline with the complete one. It never imputes zero, replaces a complete baseline, or accepts an intervening commit. The final witness still requires one rollback, increased undo, unchanged

commits and transaction end. The frozen source snapshot prevents development edits changing a running campaign's implementation. See the observer diagnosis.

Preserved earlier declarations

Declaration	Outcome retained
Equipment-01	First reference returned insufficient-evidence; 55 slots unstarted. Rollback and retry commit coalesced in the sampling interval.
Equipment-02	Three positive passes, then an operator-induced source-drift execution failure; 52 slots unstarted. Its fourth publication fails the unchanged implementation replay check.
Equipment-03	One positive pass, then insufficient-evidence; 60 slots unstarted. The observer retained an incomplete Oracle baseline despite a later admissible complete sample.
Equipment-04	New frozen declaration; all 62 fresh checks and publications passed, followed by source-review validation and acceptance.

Post-rollback observation windows, execution snapshots and the v4 observer were introduced in new declarations. No prior failure was overwritten, resumed or relabelled passed. A development reassessment of older captures is not new native qualification.

Declared limits

- Lock-duration and duplicate-retry mutants are not scored. A native unique business-key index prevents the proposed duplicate retry; prevention is not counted as a judge rejection. Duplicate invoice-line cardinality is tested.
- The rollback mutation models a leaked committed draft; it does not disable the database's rollback implementation.
- The timestamp mutation is a paired comparison rejection using distinct native second offsets. The accepted timestamp exception remains narrowly scoped.
- Operator signatures establish provenance, not independent third-party attestation. Human review records preserve the operator's explicit statements.
- No production qualification, unfamiliar-journey performance or autonomous generation rate is established by Stage A.

Stage B execution and qualified bindings

PR #223 merged Stage A at `fba89df525f2994998480172fa8aa2568da84a74`. It retained the historical `tools.ms94_controller` prototype, which expects 56 results and earlier components. That prototype is not the running entry point. The later Stage B integration is on `codex/ms94-stage-b-measurement` (prepared in commit `2ff87fe`), and is not part of that merged Stage A commit. Source integration and regression checks do not constitute a new human source review.

The running immutable snapshot uses `tools.ms94_measure_v4` and `tools.ms94_controller_v4`, validated against the real signed equipment-04:

Binding	Actual running component
Stage A admission	62 results, accepted plan, both human reviews and published inventory
Native gate	`tools.ms94_v4_gate` -> `lightyear_calibration.ms94_v4_judge`
Judge version	`idempiere-qualified-judge-v5-observer-ms94`
Observer	`qualification_observer_runtime_v4.Observer` and v4 capture verification
Support	`qualification-ms94-v3/public/JourneySupport.java`
Public shapes	`qualification-ms94-v3/public/operations-shapes.json`

All 488 qualified implementation hashes match the accepted equipment. A residual unused import of the v3 `result` helper in the native adapter does not select the judge: its evaluation function is the qualified v5 gate. Tests assert the actual function/class identities, not just version labels. The running frozen code is not edited by this documentation or the added regression tests.

Stage B 01 stopped as invalid-provenance because its strict event validator rejected a CLI deprecation warning. It remains void, with one builder call, one compilation, no native attempts and twelve unstarted slots. Following explicit user authorization to fix and restart, Stage B 02 was frozen separately with the canonical `multi_agent` flag and narrowly defined diagnostic handling.

Stage B 02 plan fingerprint: `040824a01d74dae376fd9986b3abaa5ceacbac7c6ff0e3013c42ae12c831aca0`. Snapshot fingerprint: `d5119b86a622347f962949c23e6cec54be90d525861289b0f372037da7a36da3`. The plan, declaration and manifest were publicly pushed in commit `4bc6f91` after pilots began. This is explicitly not a pre-start external timestamp. No measured cohort trial had started at the publication audit. Pilots remain outside the measurement denominator.

The authorized workflow is three pilots followed by ten fresh baseline operations trials, at most 65 calls, 39 compilations and 26 hours. Controller/provenance errors, judge-error or insufficient-evidence void the measurement; slots are never replaced. Outcomes and costs remain separate for the two declarations. Stage B has now completed that cohort; the recorded rate and adjudication limits are above.

Stage B 01's separate void outcome consumed 293,383 input tokens (249,600 cached), 12,725 output tokens, 444.141 agent seconds and 459.047 total seconds. Its cost is not included in the Stage B 02 table, and its invalid outcome is not repaired by the later declaration.

Provenance of public inputs and support

Input or component	Earlier evidence that informed it
Public model/workflow API example	MS86 boundary journey, with assertions removed before use as builder input
Typed Posted accessor and explicit posting guard	MS88/MS89 representation and reposting failures
Public verified method signatures	MS89 inaccessible methods and wrong declaring-class repairs
Plain decimal serialization and null token	MS89/MS90 comparison and trace-format failures

Input or component	Earlier evidence that informed it
Exact numeric comparison by column type; owned timestamp scope	MS90/MS91 rule registration and datatype inventory
Aggregate order-linked invoice requirement	MS92 shipment-linked shapes rejected by the private judge; made explicit in MS93/MS94
Purchasing costing/history predicates	MS92 observed purchasing effects, investigated against pinned application source in MS93
JourneySupport cleanup, trace and posting plumbing	MS93 deterministic component and isolated reference checks
v3 Oracle rollback observer	MS93 missing counter and immediate-retry sampling failures
Post-rollback observation window	The first stopped MS94 positive control; approved in equipment-02, whose first three operations controls passed
Typed errors, stricter support and purchasing ownership	MS94 technical pre-review, conditionally accepted for resolution by Howard Weale
Separate execution-source snapshots	Equipment-02 slot four exposed per-attempt discovery of development files; regression now checks copy isolation and a closed module inventory

These inputs make any future first-try result a result with inherited lessons, not a claim that the builder was unfamiliar with the workflow.

Evidence and verification

Full captures, signed records and replay instructions include all 62 complete archives as GitHub release assets, their signed archive receipts, the exact frozen replay source, human approvals and the terminal audit. The unchanged verifier reproduced every complete gate, including the intended negative verdicts. The focused deterministic regression suite passed 166 tests.

Frozen plan: `77b075d0cafebe41dc33df02e43150a18a72e4601750ca3faa0211884960257e`. Acceptance: `0ad73e79fa94f7e74949e0733d655d814c9852c48fd9333643648d1b240480b4`. Terminal audit: `42c57d635c4336c0375c328918caaba9d25a94d13a2bf3b53b29d516e9923c44`.

Prospective retention-rule qualification and Stage B 03

Howard approved the order-derived invoice-type rule as the single next intervention. Its fresh additive qualification completed in 23 minutes 38.844 seconds with zero model calls: two positive passes and one expected negative rejection, all publications replayed and cleanup verified. All 488 accepted equipment-04 implementation files remain unchanged. The new construction variants have no independent human source attestation; the inherited reference was reviewed and the prospective rule was user-approved.

Stage B 03 is separately frozen for three excluded pilots and 20 fresh cohort trials, capped at 115 calls, 69 compilations and 26 hours (five calls/three compiles per trial). Its plan is `f613b60c3515fa31d79f14929d7a2adb805641f4866fb5595d023d421ea7d41f`. Its declaration was published before generation, and the one-shot launch record binds the confirmed public commit. No smoke tool or wider diagnostics were added. The original 2/10 interval and adjudication caveats above remain unchanged; these qualification controls are not autonomous successes.

Stage B 03 launched once at 19:33:05 UTC on 29 September 2026, after remote commit `026fdf804fe27953cf41c8f9552f66476bff3ec9` was confirmed. The launch and signed publication proof record this ordering. The completed Stage B 03 result and verification are reported above.

B04 prospective freeze

Stage B 04 prepares one new 23-slot closed-runtime-diagnostic campaign after all qualification prerequisites passed. The snapshot retains all 946 paired-A3 inputs unchanged. Fifty focused DEV tests and 34 included frozen-root tests passed. The initial builder inputs match B03; pooled n=40 reporting is explicitly omitted because native assessment dates differ. No B04 model calls or native attempts have occurred. Exact public source, plans, declarations and allowlists must be verified before the one authorized launch.